

watt terrace and backyard plants is less than two months.

6. Solar technologies are stabilised and proven.

7. There is no need to store power if all is delivered to grid to meet large daytime demand.

8. The cost per watt of PV modules is progressively dropping – 50 per cent drop during the last two years.

9. Many technology initiatives are being taken at various stages to increase efficiency and reduce cost.

10. There is a scope to encourage and create profitable SMEs generating 100-500kWp PV solar power into grid at substation level at 400V three-phase.

Shortcomings in the current plan

Currently we are broadly following the western model for utilisation of solar power to feed into the grid. Government encourages establishing over one megawatt-peak (MWp) solar plants with a special preference to even larger plants of 10 or 20 MWp. This approach suffers from many problems:

1. All PV solar plants of even 10MW capacity convert DC solar power into grid-synchronised AC power with inverters of 500 kilowatt capacity. All this power is generated at 400V three-phase with 50Hz frequency. In large plants, output of several inverters at 400V is boosted to 33kV or 66kV and fed into the grid. Grid then steps this down to 400V at the consumer end. This uses several distribution transformers resulting in the loss of 12 to 15 per cent of expensive power in transformer losses alone.

2. At 33kV and above, at any given moment there is hundreds of gigawatt of floating power. A few megawatt pumped into that huge pool is a minor addition with no significant advantage to the final consumer. It will bring no measurable relief in power problems at the consumer end.

However, if the same power is injected at the power-starved high-impedance end of the grid during daytime, it will instantly eliminate the

blackout and prevent wild swings in that service area. One has to remember that energy injected will first serve the local customers since it can't be fed into higher grid unless the local needs are fully met.

3. Each megawatt plant needs about 12,140 square metre (3 acre) of land. Since basic module of 1.8 square metre generates only about 250Wp, one needs to connect several thousand of these to form an array. A 10-megawatt plant needs around 101,171-121,406 square metre (25 to 30 acre). We have millions of square kilometre unused terraces and flat roofs all over the country. These can be used since investment needed to set up smaller plants on terraces is the same per kWp for a 100 kilowatt or a 10 megawatt plant. In fact it can be cheaper for smaller plants.

4. Current architecture of solar arrays is prone to large losses besides being complex in their network formation in terms of junction boxes, etc.

(i) Since each solar module generates 30 to 40V DC, one needs to connect them in series strings to build voltage to about 500V peak. Hundreds of such series strings are connected in parallel to build about 500 to 600kWp power. This leads to complexity and the need to use matched panels of identical design.

(ii) Modules connected in series strings often lose full energy output even if a shadow or dirt-like bird dropping cover just a part of one module in a string. This loss of harvest from entire string could be as high as 75 per cent.

(iii) Maximum peak power tracking (MPPT) of such a complex matrix of modules leads to many of its modules delivering less power than its peak capacity, leading to poor energy harvesting.

(iv) Since module currents are 6 to 8A, complex wiring leads to high resistive losses. The latest technology therefore uses all parallel-connected modules with built-in DC-to-DC energy maximisers or optimisers. Each module delivers its maximum power point at a high voltage and thus fa-

cilitates connecting all PV modules in an array in parallel and deliver their energy in a common bus bar.

Towards solar power

Giving importance to solar-generated power has lot of benefits. Improvement in power quality, minimising voltage swings within limits and elimination of blackout for consumers at the terminal ends of the grid can be best achieved by injecting 50 to 250 kilowatt PV solar energy at 400V substation level. The injected power will feed the local loads and only the excess locally unused power will get stepped up into the grid. Same power, if fed into grid at 11 or 33kV will not help improving the local power problems, since it will be a very small fraction of the power flowing through grid at that potential.

The power shortage on the grid is primarily during the daytime. Therefore solar power is ideal to meet that shortfall. This will induce large increase in productivity due to availability of stable reliable power to local communities. It will eliminate unnecessary investment for the consumers in diesel generators or inverters and batteries.

Daytime injection of solar power at a local electrical substation will reduce the heat in the transformers and thus increase their life. Investment in such small power companies will be same in terms of the capital. Cost per kilowatt will be the same, if not less, than in the solar plants of megawatt size. These power SMEs will maintain plants more carefully than the large plants and will also result in growth of local employment opportunities. Use of all parallel array architecture in such plants will lead to substantial increase in power harvesting than the conventional series and parallel complex matrix presently used. Figures could be as high as 20 per cent. ●

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